

PASTA worksheet

Stages	Sneaker company
I. Define business and security objectives	Make 2-3 notes of specific business requirements that will be analyzed. • <i>Will the app process transactions?</i> • <i>Does it do a lot of back-end processing?</i> • <i>Are there industry regulations that need to be considered?</i> 1. <i>Ensure secure and seamless transactions between buyers and sellers through the mobile app.</i> 2. <i>Protect user account data and payment information in compliance with industry privacy standards.</i> 3. <i>Maintain app reliability and uptime to support real-time sneaker listings and purchases.</i>
II. Define the technical scope	List of technologies used by the application: • <i>Application programming interface (API)</i> • <i>Public key infrastructure (PKI)</i> • <i>SHA-256</i> • <i>SQL</i> Write 2-3 sentences (40-60 words) that describe why you choose to prioritize that technology over the others. I will prioritize SQL because it directly handles sensitive user and transaction data stored in the database. SQL queries are critical for retrieving sneaker listings and processing purchases, making them a high-risk area for injection attacks and data breaches. Securing SQL operations ensures data integrity and prevents unauthorized access.
III. Decompose application	<u>Sample data flow diagram</u> The user initiates a sneaker search. The app queries the database for listings. Results are returned and displayed to the user. This process highlights how user input interacts with backend data,

	identifying potential exposure points for injection or data manipulation.
IV. Threat analysis	List 2 types of threats in the PASTA worksheet that are risks to the information being handled by the application. • <i>What are the internal threats?</i> SQL injection through poorly validated input fields. Misconfigured API endpoints are exposing internal data. • <i>What are the external threats?</i> Session hijacking via insecure cookies or weak encryption. Credential stuffing attacks exploit weak passwords.
V. Vulnerability analysis	List 2 vulnerabilities in the PASTA worksheet that could be exploited. • <i>Could there be things wrong with the codebase?</i> • <i>Could there be weaknesses in the database?</i> • <i>Could there be flaws in the network?</i> 1. <i>Lack of input validation and parameterized queries in SQL statements.</i> 2. <i>Weak session management allows attackers to reuse stolen tokens.</i>
VI. Attack modeling	Sample attack tree diagram Attacker Goal: Compromise user data. Paths: • Exploit SQL injection to extract database records. • Hijack user sessions through insecure tokens. • Use weak credentials to gain unauthorized access.
VII. Risk analysis and impact	List 4 security controls that you've learned about that can reduce risk. 1. Implement parameterized queries and strict input validation to prevent SQL injection. 2. Enforce multi-factor authentication (MFA) for all user accounts. 3. Use secure session management with HTTPS and

	short-lived tokens. 4. Apply continuous monitoring and logging to detect suspicious activity.
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